ZHAOHUA CHEN

Taobao & Tmall Group of Alibaba,
Chaoyang, Beijing, China
Gaoling School of Artificial Intelligence, Renmin University of China,
Haidian, Beijing, China
Email: chenzhaohua.czh@alibaba-inc.com, caezoah@gmail.com, chenzhaohua@pku.edu.cn
Homepage: https://daleczh.github.io/

RESEARCH INTERESTS

I am mainly working on the (complicated) game-theoretic interaction between the servicing platform and served targets (e.g., advertising platform v.s. advertisers, e-commerce platform v.s. sellers, platform v.s. users, etc.), mainly in the following aspects:

- How can the platform design algorithms to reach its goal or help targets reach their goals?
- How can the targets conversely exploit the platform's algorithm to better benefit themselves?
- How can one simultaneously benefit the platform and their targets?

I also work on various topics in the interplay of computer science and economics, known as algorithmic game theory.

PROFESSIONAL EXPERIENCE

Algorithm Expert 2026.08 –
Taobao & Tmall Group of Alibaba, Chaoyang, Beijing, China

Postdoctoral Researcher 2026.08 –
Taobao & Tmall Group of Alibaba, Chaoyang, Beijing, China
Renmin University of China, Haidian, Beijing, China
Gaoling School of Artificial Intelligence
Advisor: Jian Xu, Qi Qi

EDUCATION

Ph.D. of Science 2021.09 – 2026.07
Peking University, Haidian, Beijing, China
Center on Frontiers of Computing Studies (CFCS), School of Computer Science
Advisor: Xiaotie Deng

Bachelor of Science, Summa Cum Laude 2017.09 – 2021.07
Peking University, Haidian, Beijing, China
Turing class of AI
Intelligence Science and Technology
School of Electronics Engineering and Computer Science
GPA: 3.768 / 4

PUBLICATIONS

WORKING PAPERS

[P1] **Advancing Auto-bidding with Theory-Empowered LLMs**
Qingmao Yao, Qun Hu, **Zhaohua Chen**, Xinyu Zhang, Shiping Song, Qianyu Zhu, Yang Tan, Dagui Chen, Jian Xu, Xiaotie Deng

[P2] **Online Resource Allocation with an Endogeneous Markov State: Fewer LP Solves Earn More**
Zhaohua Chen
arXiv: 2610.09577

[P3] **LERA: LLM-Enhanced RAG for Ad Auction in Generative Chatbots**
Haoran Sun, Xinrui Song, Xinyu Zhang, **Zhaohua Chen**, Xu Chu, Zhilin Zhang, Chuan Yu, Jian Xu, Bo Zheng, Xiaotie Deng
ICML 2026 Workshop on New Frontiers in Game-Theoretic Learning (NExT-GAME)
arXiv: 2605.16474

[P4] **Pareto-Efficient Multi-Buyer Mechanisms: Characterization, Fairness and Welfare**
($\alpha\beta$) Moshe Babaioff, Sijin Chen, **Zhaohua Chen**, and Yiding Feng
EAAMO 2026
arXiv: 2602.11967

[P5] **Budget-Constrained Auctions with Unassured Reports: Strategic Equivalence and Structural Properties**
Zhaohua Chen, Mingwei Yang, Chang Wang, Jicheng Li, and Xiaotie Deng
Major revision at ***Management Science***
EC 2025 Workshop on Frontiers of Online Advertising (Spotlight)
Preliminary conference version appeared in WWW 2024

JOURNAL ARTICLES

[J1] **On Tightness of Tsaknakis-Spirakis Descent Methods for Approximate Nash Equilibria**
($\alpha\beta$) **Zhaohua Chen**, Xiaotie Deng, Wenhan Huang, Hanyu Li, and Yuhao Li
Information and Computation
doi: 10.1016/j.ic.2023.105046

[J2] **An Efficient and Robust Committee Structure for Sharding Blockchain**
Mengqian Zhang*, Jichen Li*, **Zhaohua Chen***, Hongyin Chen*, and Xiaotie Deng
IEEE Transactions on Cloud Computing
doi: 10.1109/TCC.2022.3217856

[J3] **A Provable Softmax Reputation-based Protocol for Permissioned Blockchains**
($\alpha\beta$) Hongyin Chen*, **Zhaohua Chen***, Yukun Cheng, Xiaotie Deng, Wenhan Huang, Jichen Li, Hongyi Ling*, and Mengqian Zhang
IEEE Transactions on Cloud Computing
doi: 10.1109/TCC.2021.3130244

CONFERENCE PROCEEDINGS

[C1] **Ad Insertion in LLM-generated Responses**
Shengwei Xu*, **Zhaohua Chen***, Xiaotie Deng, Zhiyi Huang, and Grant Schoenebeck
Workshop on Agentic and Generative AI for E-Commerce (GenAI-eCom) 2026 (Oral)
Marketplace Innovation Workshop (MIW) 2026
EC 2025 Workshop on Frontiers of Online Advertising (Spotlight)
arXiv: 2601.19435

[C2] **Contextual Decision-Making with Knapsacks Beyond the Worst Case**
Zhaohua Chen, Rui Ai, Mingwei Yang, Yuqi Pan, Chang Wang, and Xiaotie Deng
NeurIPS 2024
arXiv: 2211.13952

[C3] **Are Bounded Contracts Learnable and Approximately Optimal?**
($\alpha\beta$) Yurong Chen, **Zhaohua Chen**, Xiaotie Deng, and Zhiyi Huang
EC 2024
EC 2025 Workshop on Online Learning and Economics
doi: 10.1145/3670865.3673483
arXiv: 2402.14486

[C4] **Robust Decision Aggregation with Second-order Information**
Yuqi Pan, **Zhaohua Chen**, and Yuqing Kong
WWW 2024
doi: 10.1145/3589334.3645384
arXiv: 2311.14094

[C5] **Budget-Constrained Auctions with Unassured Priors: Strategic Equivalence and Structural Properties**
Zhaohua Chen, Mingwei Yang, Chang Wang, Jicheng Li, Zheng Cai, Yukun Ren, Zhihua Zhu, and Xiaotie Deng
WWW 2024 (Oral)
doi: 10.1145/3589334.3645344
arXiv: 2203.16816
SSRN: 4071291

[C6] **Dynamic Budget Throttling in Repeated Second-Price Auctions**
Zhaohua Chen*, Chang Wang*, Qian Wang*, Yuqi Pan, Zhuming Shi, Zheng Cai, Yukun Ren, Zhihua Zhu, and Xiaotie Deng
AAAI 2024
doi: 10.1609/aaai.v38i9.28816
arXiv: 2207.04690

[C7] **Coordinated Dynamic Bidding in Repeated Second-Price Auctions with Budgets**
Yurong Chen*, Qian Wang*, Zhijian Duan, Haoran Sun, **Zhaohua Chen**, Xiang Yan, and Xiaotie Deng
ICML 2023
arXiv: 2306.07709

[C8] **Decentralized Asset Custody Scheme with Security against Rational Adversary**
($\alpha\beta$) **Zhaohua Chen** and Guang Yang
WINE 2021
doi: 10.1007/978-3-030-94676-0_25

arXiv: 2008.10895
Cryptology ePrint Archive: 2020/1229

[C9] **On Tightness of the Tsaknakis-Spirakis Algorithm for Approximate Nash Equilibrium**
($\alpha\beta$) **Zhaohua Chen**, Xiaotie Deng, Wenhan Huang, Hanyu Li*, and Yuhao Li
SAGT 2021
doi: 10.1007/978-3-030-85947-3_7
arXiv: 2107.01471

[C10] **An Efficient Permissioned Blockchain Model with Provable Reputation Mechanism**
($\alpha\beta$) Hongyin Chen*, **Zhaohua Chen***, Yukun Cheng, Xiaotie Deng, Wenhan Huang, Jichen Li, Hongyi Ling*, and Mengqian Zhang
ICDCS 2021 *(Poster)*
doi: 10.1109/ICDCS51616.2021.00123
arXiv: 2002.06852

[C11] **CycLedger: A Scalable and Secure Parallel Protocol for Distributed Ledger via Sharding**
Mengqian Zhang*, Jichen Li*, **Zhaohua Chen***, Hongyin Chen*, and Xiaotie Deng
IPDPS 2020 *(Best paper nominee)*
doi: 10.1109/IPDPS47924.2020.00045
arXiv: 2001.06778

PATENTS

1. CN112419060B. **Asset hosting system, asset management method, node and medium**
Inventor: YANG GUANG, **CHEN ZHAOHUA**, WU MING, LONG FAN, LI CHENXING

INTERNSHIPS

Research Assistant at **The University of Hong Kong** 2025.05 – 2025.10
Mentor: Zhiyi Huang

Research Intern at **Taobao & Tmall Group of Alibaba** 2024.07 – 2026.07
Mentor: Zhilin Zhang
Topic: Design and analysis of mechanisms and algorithms in the e-commerce market

Research Intern at **DeCus.io (under Conflux)** 2020.09 – 2021.10
Mentors: Ge Chen, Tian Lin, Guang Yang
Topic: Group assignment scheme design in decentralized asset custody

Research Intern at **Conflux** 2020.03 – 2020.09
Mentor: Guang Yang
Topic: Mechanism design and analysis in blockchain cross-chain protocols

TEACHINGS

Advisor:

Billiards	Yuanpei College, Peking University, 2023.03 – 2023.06
Billiards	Yuanpei College, Peking University, 2022.10 – 2022.12

Teaching Assistant:

Discrete Math and Structures (I)	Peking University, 2023.09 – 2024.01
Study and Practice on Topics of Frontier Computing (II)	Peking University, 2023.02 – 2023.06
Study and Practice on Topics of Frontier Computing (I)	Peking University, 2022.09 – 2023.01
Discrete Math and Structures (I)	Peking University, 2021.09 – 2022.01
Discrete Math and Structures (I)	Peking University, 2020.09 – 2021.01
Blockchain Theory and Technique	Peking University, 2019.09 – 2020.01

SERVICES

Journal Reviewing:
SIAM Journal on Computing

Conference Reviewing:
EC 2026, FOCS 2024, GameSec 2021, ICDCS 2022, ICLR 2026, ICML 2025, NeurIPS {2025, 2024}, WINE {2026, 2025}, WSDM 2026, WWW 2024

Services in Academic Organizations:

Committee member	CCF TCCE, 2026.09 –
Administrative assistant	CCF TCCE, 2022.08 – 2026.08

Organizing:

Chair of undergraduate research forum	IJTCS 2021
Leader of volunteer group	WINE 2020

Volunteering:
IJTCS-FAW 2023, CSIAM-BTAF 2021, IJTCS 2020

HONORS & AWARDS

Excellent Doctoral Dissertation of School of Computer Science, Peking University	2026
Excellent (Ph.D.) Graduate of Peking University	2026
Short-term Exchange Program for Ph.D. Students in Peking University	2024
Excellent (B.Sc.) Graduate of Peking University	2021

3rd Place in Inter-City College Student Billiards Invitational Tournament (Team of PKU)	2025
3rd Place in Captial College Student Billiards Championship (Single)	2021
2nd Place in Captial College Student Billiards Championship (Team of PKU)	2019